

Systems of First Order Linear ODEs

1 First Order ODE Systems

1.1 Non-Linear and Linear Systems

The general system of first order equations involving n dependent variables defined on an interval $I \subseteq \mathbb{R}$ can be written as

$$\begin{cases} y_1'(t) = F_1(t, y_1, \dots, y_n) \\ y_2'(t) = F_2(t, y_1, \dots, y_n) \\ \vdots \\ y_n'(t) = F_n(t, y_1, \dots, y_n) \end{cases} \quad \text{for } t \in I. \quad (1)$$

with initial conditions $y_1(t_0) = x_1, \quad y_2(t_0) = x_2, \quad \dots, \quad y_n(t_0) = x_n,$

The solution to the system can be expressed in vector form as

$$\mathbf{y}(t) = \begin{pmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_n(t) \end{pmatrix}$$

Theorem 1.1 (Existence and Uniqueness).

For the system of first order ODEs (1), if the functions $F_1, F_2, \dots, F_n$ and the partial derivatives $\frac{\partial F_i}{\partial y_j}$ for all pairs $i, j = 1, 2, \dots, n$ are continuous on a region R defined by

$$R := [t_0 - a, t_0 + a] \times [x_1 - b, x_1 + b] \times \dots \times [x_n - b, x_n + b] \subseteq \mathbb{R}^{n+1},$$

where constants $a, b > 0$, then there exists a unique solution $\mathbf{y}(t)$ for $t \in (t_0 - h, t_0 + h)$ to the IVP, where

$$h = \min\{a, b/M\} \quad \text{and} \quad M = \max_{\mathbf{v} \in R} \{|F_1(\mathbf{v})|, |F_2(\mathbf{v})|, \dots, |F_n(\mathbf{v})|\}.$$

Any n -th order ODE can be rewritten as a system of n first order ODEs. Given an n -th order ODE

$$y^{(n)} = F(t, y, y', \dots, y^{(n-1)}),$$

setting $z_1 = y, z_2 = y', \dots, z_n = y^{(n-1)}$, we obtain the system

$$\begin{cases} z_1' = z_2 \\ z_2' = z_3 \\ \vdots \\ z_{n-1}' = z_n \\ z_n' = F(t, z_1, z_2, \dots, z_n) \end{cases}$$

which is a system of n first order ODEs.

Definition 1.1 (First-Order Linear System).

A system of first order ODEs is called **linear** if it can be expressed in the form

$$\mathbf{y}'(t) = \mathbf{P}(t)\mathbf{y}(t) + \mathbf{g}(t) \quad \text{for } t \in I, \quad (2)$$

where

$$\mathbf{y}(t) = \begin{pmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_n(t) \end{pmatrix} \quad \mathbf{g}(t) = \begin{pmatrix} g_1(t) \\ g_2(t) \\ \vdots \\ g_n(t) \end{pmatrix} \quad \mathbf{P}(t) = \begin{pmatrix} p_{11}(t) & p_{12}(t) & \cdots & p_{1n}(t) \\ p_{21}(t) & p_{22}(t) & \cdots & p_{2n}(t) \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1}(t) & p_{n2}(t) & \cdots & p_{nn}(t) \end{pmatrix}$$

Definition 1.2 (Homogeneous First-Order Linear System).

A first order linear system is called **homogeneous** if $\mathbf{g}(t) = \mathbf{0}_n$ for all $t \in I$, i.e., it can be expressed as

$$\mathbf{y}'(t) = \mathbf{P}(t)\mathbf{y}(t) \quad \text{for } t \in I.$$

Theorem 1.2 (Existence and Uniqueness for First-Order Linear Systems).

For the first-order linear system (2), if all entries of $\mathbf{P}(t)$ and $\mathbf{g}(t)$ are continuous on an interval I , then for any $t_0 \in I$ and $\mathbf{y}_0 \in \mathbb{R}^n$, there exists a unique solution $\mathbf{y}(t)$ to the IVP

$$\begin{cases} \mathbf{y}'(t) = \mathbf{P}(t)\mathbf{y}(t) + \mathbf{g}(t) \\ \mathbf{y}(t_0) = \mathbf{a} \in \mathbb{R}^n \end{cases} \quad \text{for } t \in I.$$

1.2 Homogeneous Linear Systems

In this section, we will focus on the following homogeneous system defined on an open interval $I \subseteq \mathbb{R}$:

$$\mathbf{y}'(t) = \mathbf{P}(t)\mathbf{y}(t) \quad \text{for } t \in I, \quad (3)$$

where $\mathbf{y} \in C^1(I)^n$ and $\mathbf{P} \in C(I)^{n \times n}$, i.e., all entries of $\mathbf{P}$ and $\mathbf{y}$ are continuous on I .

Theorem 1.3 (Principle of Superposition).

If $\mathbf{y}_1(t)$ and $\mathbf{y}_2(t)$ are solutions to the homogeneous first-order linear system (3), then the linear combination

$$\mathbf{y}(t) = c_1\mathbf{y}_1(t) + c_2\mathbf{y}_2(t)$$

is also a solution for any constants c_1, c_2 .

Definition 1.3 (Wronskian).

The **Wronskian** of n vector functions $\mathbf{f}_1(t), \mathbf{f}_2(t), \dots, \mathbf{f}_n(t) \in C(I)^n$ is defined as

$$W(\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_n)(t) := \det \left(\begin{bmatrix} | & & | \\ \mathbf{f}_1(t) & \cdots & \mathbf{f}_n(t) \\ | & & | \end{bmatrix} \right) \quad \text{for } t \in I.$$

Remark. The Wronskian for vector functions is different from the Wronskian for scalar functions. There is no derivative involved in the definition of the Wronskian for vector functions.

For any $t \in I$,

$$W(\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_n)(t) \neq 0 \iff \mathbf{f}_1(t), \mathbf{f}_2(t), \dots, \mathbf{f}_n(t) \text{ are linearly independent.}$$

Theorem 1.4 (Liouville's Formula).

Let $\mathbf{y}_1(t), \mathbf{y}_2(t), \dots, \mathbf{y}_n(t)$ be n solutions to the homogeneous system (3). Then the Wronskian of these solutions is given by

$$W(\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n)(t) = \mathbf{c} \cdot \exp\left(\int \text{tr}(\mathbf{P}(t)) dt\right) \quad \text{for } t \in I,$$

where $\mathbf{c}$ is a constant not depending on t .

Remark. The **trace** of a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is defined as the sum of its diagonal entries:

$$\text{tr}(\mathbf{A}) := a_{11} + a_{22} + \dots + a_{nn}.$$

Proof (Liouville's Formula).

Before the proof, we need to introduce several theorems from linear algebra:

1. For any square matrix $\mathbf{A}$,

$$\frac{d}{dt}(\det(\mathbf{A})) = \text{tr}\left(\text{adj}(\mathbf{A}) \frac{d\mathbf{A}}{dt}\right). \quad (4)$$

2. The trace remains unchanged under cycle permutations:

$$\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{BCA}). \quad (5)$$

3. Property of the adjugate matrix:

$$\text{adj}(\mathbf{A})\mathbf{A} = \mathbf{A} \text{adj}(\mathbf{A}) = \det(\mathbf{A})\mathbf{I}. \quad (6)$$

Now, let $\mathbf{Y} := \begin{bmatrix} \mathbf{y}_1 & \mathbf{y}_2 & \dots & \mathbf{y}_n \end{bmatrix} \in C^1(I)^{n \times n}$ and $W := \det(\mathbf{Y})$ be the Wronskian. Then we have

$$\begin{aligned} W' &= \text{tr}(\text{adj}(\mathbf{Y})\mathbf{Y}') \\ &= \text{tr}(\text{adj}(\mathbf{Y})\mathbf{PY}) \quad \text{since } \mathbf{y}' = \mathbf{Py} \\ &= \text{tr}(\mathbf{Y} \text{adj}(\mathbf{Y})\mathbf{P}) \\ &= \text{tr}(\mathbf{WIP}) \\ &= W \text{tr}(\mathbf{P}). \end{aligned}$$

Hence the Liouville's formula. □

Theorem 1.5.

If $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n$ are n solutions to a homogeneous system (3), then

$$\forall t \in I, \quad W(\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n)(t) = 0 \iff \exists t_0 \in I \text{ s.t. } W(\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n)(t_0) = 0.$$

Proof.

Proven by Liouville's formula. □

Theorem 1.6.

If the n solutions $\mathbf{y}_1(t), \mathbf{y}_2(t), \dots, \mathbf{y}_n(t)$ of the homogeneous system (3) are **linearly independent** for each $t \in I$, then every solution of the system can be expressed as a linear combination:

$$\mathbf{y}(t) = c_1 \mathbf{y}_1(t) + c_2 \mathbf{y}_2(t) + \dots + c_n \mathbf{y}_n(t) \quad \text{for } t \in I$$

in **exactly one way**.

Remark. The Principle of Superposition indicates the solutions form a vector space, thus there exist bases. Therefore, the essence of this theorem is to show that the dimension of the solution space is exactly n .

Proof.

Let $\phi(t)$ be any solution to the homogeneous system and let $\mathbf{Y} := [\mathbf{y}_1 \quad \mathbf{y}_2 \quad \dots \quad \mathbf{y}_n]$. We pick some $t_0 \in I$ and define the constant vector $\xi := \phi(t_0) \in \mathbb{R}^n$. Since $\mathbf{y}_1(t), \dots, \mathbf{y}_n(t)$ are linearly independent for any $t \in I$, we have $\det(\mathbf{Y})(t_0) \neq 0$. Then the following equation has a unique solution $\mathbf{c} \in \mathbb{R}^n$:

$$\mathbf{Y}(t_0) \cdot \mathbf{c} = \xi \quad \text{for } \mathbf{c} \in \mathbb{R}^n.$$

Now we define the linear combination of $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n$ as a new vector function:

$$\eta(t) := \mathbf{Y}(t) \cdot \mathbf{c} \quad \text{for } t \in I.$$

Construct the following IVP: the homogeneous system (3) with the initial condition $\mathbf{y}(t_0) = \xi$. Since $\phi(t)$ and $\eta(t)$ are both solutions to this IVP, by the uniqueness of solutions (Theorem 1.2), we have $\phi(t) = \eta(t)$ for all $t \in I$. Thus any solution $\phi(t)$ can be expressed as a linear combination of the fundamental solutions. □

Definition 1.4 (Fundamental Set of Solutions).

A set of n **linearly independent** solutions $\{\mathbf{y}_1(t), \mathbf{y}_2(t), \dots, \mathbf{y}_n(t)\}$ to the homogeneous system (3) is called a **fundamental set of solutions**.

Definition 1.5 (Fundamental Matrix).

A matrix function $\mathbf{Y}(t) \in C^1(I)^{n \times n}$ is called a **fundamental matrix** of the homogeneous system (3) if its columns form a fundamental set of solutions.

For a square matrix $\mathbf{Y}$ whose columns are solutions to the homogeneous system (3), we have

$$\det(\mathbf{Y}) \neq 0 \quad \Longleftrightarrow \quad \mathbf{Y} \text{ is a fundamental matrix.}$$

Theorem 1.7 (Existence of Fundamental Matrix).

For the homogeneous system (3), there exists at least one fundamental matrix $\mathbf{Y}(t)$.

Proof.

We need to show that there exist at least n linearly independent solutions. We can obtain the solutions with Theorem 1.2 by constructing n IVPs with initial conditions at the same point t_0 but different values.

For the i -th IVP, we set the initial condition as $\mathbf{y}(t_0) = \mathbf{e}_i$, where $\mathbf{e}_i$ is the i -th standard basis vector. Then $\mathbf{Y}(t_0) = \mathbf{I}_n$. Since the Wronskian $W(\mathbf{y}_1, \dots, \mathbf{y}_n)(t_0) = \det(\mathbf{Y})(t_0) = 1 \neq 0$, the solutions are linearly independent throughout I by Theorem 1.5. $\square$

Theorem 1.8.

Consider the homogeneous system (3) with $\mathbf{P} \in C(I, \mathbb{R}^{n \times n})$. Suppose $\mathbf{u}, \mathbf{v} \in C(I, \mathbb{R}^n)$ are two vector functions such that $\mathbf{y} = \mathbf{u} + i\mathbf{v} \in C(I, \mathbb{C}^n)$ is a solution to the system. Then both $\mathbf{u}$ and $\mathbf{v}$ are solutions to the system.

2 Homogeneous Systems with Constant Coefficients

In this section, we will focus on the following system defined on an open interval $I \subseteq \mathbb{R}$:

$$\mathbf{y}'(t) = \mathbf{A} \cdot \mathbf{y}(t) \quad \text{for } t \in I, \quad (7)$$

where $\mathbf{A} \in \mathbb{R}^{n \times n}$ is a constant matrix. We will introduce two methods to solve this system:

1. The method of matrix exponentials.
2. The method of eigenvalues and eigenvectors.

2.1 Method 1: Matrix Exponentials

2.1.1 Matrix Exponentials

Theorem 2.1.

For any square matrices $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$,

$$\mathbf{AB} = \mathbf{BA} \implies e^{\mathbf{A}+\mathbf{B}} = e^{\mathbf{A}}e^{\mathbf{B}}.$$

Proof.

Using the Binomial Theorem. $\square$

Theorem 2.2.

For any square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, we have $\det(e^{\mathbf{A}}) \neq 0$, and $(e^{\mathbf{A}})^{-1} = e^{-\mathbf{A}}$.

Proof.

Let $\mathbf{A}$ be an arbitrary square matrix. Since $\mathbf{A}(-\mathbf{A}) = -\mathbf{A}^2 = (-\mathbf{A})\mathbf{A}$, we have $e^{\mathbf{A}}e^{-\mathbf{A}} = e^{\mathbf{A}-\mathbf{A}} = e^{\mathbf{0}} = \mathbf{I}$. $\square$

Theorem 2.3.

For a constant square matrix $\mathbf{A}$:

$$\frac{d(e^{\mathbf{A}t})}{dt} = \mathbf{A}e^{\mathbf{A}t} = e^{\mathbf{A}t}\mathbf{A}$$

Proof.

$$\frac{d(e^{At})}{dt} = \frac{d}{dt} \sum_{k=0}^{\infty} \frac{A^k t^k}{k!} = A \sum_{k=1}^{\infty} \frac{A^{k-1} t^{k-1}}{(k-1)!} = A e^{At} = \sum_{k=1}^{\infty} \frac{A^{k-1} t^{k-1}}{(k-1)!} \cdot A = e^{At} A$$

□

Theorem 2.3 indicates that the columns of the matrix $Y(t) := e^{At}$ are solutions to the homogeneous system (7), and Theorem 2.2 shows its invertibility. Thus, $Y(t)$ is a **fundamental matrix**. Then, the **general solution** to the system can be expressed as

$$y(t) = e^{At} \cdot c \quad \text{for } t \in I,$$

where $c \in \mathbb{R}^n$ is an arbitrary constant vector.

2.1.2 Computing Matrix Exponentials

Now, how to compute e^{At} ? We can make use of the S-N Decomposition.

Definition 2.1 (Semisimple Matrix).

A square matrix $A \in \mathbb{R}^{n \times n}$ is called **semisimple** if it is **diagonalizable**, i.e., there exists an invertible matrix P such that $P^{-1}AP = D$, where D is a diagonal matrix.

Remark. The diagonal entries of D are the eigenvalues of A .

Definition 2.2 (Nilpotent Matrix).

A square matrix $A \in \mathbb{R}^{n \times n}$ is called **nilpotent** if there exists a positive integer k such that $A^k = O$.

Theorem 2.4.

For a diagonalizable matrix $A = PAP^{-1} \in \mathbb{R}^{n \times n}$, we have

$$e^A = Pe^AP^{-1} = P \begin{pmatrix} e^{\lambda_1} & & \\ & \ddots & \\ & & e^{\lambda_n} \end{pmatrix} P^{-1} \quad \text{where } \lambda_1, \dots, \lambda_n \text{ are eigenvalues of } A.$$

Theorem 2.5 (S-N Decomposition).

For any square matrix $A \in \mathbb{R}^{n \times n}$, there **uniquely** exist a semisimple matrix S and a nilpotent matrix N such that

$$A = S + N \quad \text{and} \quad SN = NS.$$

Additionally, S shares the same eigenvalues as A with the same algebraic multiplicities.

Since S and N commute, we have

$$e^{At} = e^{St+Nt} = e^{St}e^{Nt}.$$

Because S is semisimple, we can compute e^{St} with Theorem 2.4. For the nilpotent matrix N , since $N^k = O$ for some k , the series expansion of e^{Nt} terminates within finitely many terms:

$$e^{Nt} = I + Nt + \frac{N^2 t^2}{2!} + \dots + \frac{N^{k-1} t^{k-1}}{(k-1)!}.$$

Thus it is computable.

2.2 Method 2: Eigenvalues and Eigenvectors

Definition 2.3 (Generalized Eigenvector).

For a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ and an eigenvalue λ of $\mathbf{A}$, a nonzero vector $\xi \in \mathbb{R}^n$ is called a **generalized eigenvector** of $\mathbf{A}$ corresponding to λ if there exists a positive integer k such that

$$(\mathbf{A} - \lambda \mathbf{I})^k \xi = \mathbf{0}.$$

If k is the least integer satisfying the above equation, then ξ is called a generalized eigenvector of order k .

Theorem 2.6 (Generalized Eigenspace).

The **generalized eigenspace** of an eigenvalue λ of $\mathbf{A}$ with algebraic multiplicity m , denoted by G_λ , is given by

$$G_\lambda = \ker((\mathbf{A} - \lambda \mathbf{I})^m) \quad \text{and} \quad \dim(G_\lambda) = m.$$

Consider the homogeneous system (7). Suppose $\mathbf{A} \in \mathbb{R}^{n \times n}$ has k distinct eigenvalues:

$$\det(x\mathbf{I} - \mathbf{A}) = (x - \lambda_1)^{m_1}(x - \lambda_2)^{m_2} \cdots (x - \lambda_k)^{m_k},$$

where m_i is the algebraic multiplicity of λ_i and $\sum_{i=1}^k m_i = n$. For each eigenvalue λ_i , we need to find m_i linearly independent solutions: $\mathbf{y}_1^{(i)}, \mathbf{y}_2^{(i)}, \dots, \mathbf{y}_{m_i}^{(i)}$, so that we have n linearly independent solutions in total.

For each λ_i , whose geometric multiplicity is q_i , there are two cases:

1. $1 \leq q_i = m_i \leq n$, there are m_i L.I. eigenvectors;
2. $1 \leq q_i < m_i \leq n$.

Case 1

For the first case, we can construct the solutions for λ_i as

$$\begin{aligned} \mathbf{y}_1^{(i)}(t) &= \mathbf{r}_1^{(i)} e^{\lambda_i t}, \\ \mathbf{y}_2^{(i)}(t) &= \mathbf{r}_2^{(i)} e^{\lambda_i t}, \\ &\vdots \\ \mathbf{y}_{m_i}^{(i)}(t) &= \mathbf{r}_{m_i}^{(i)} e^{\lambda_i t}. \end{aligned}$$

where $\mathbf{r}_1^{(i)}, \mathbf{r}_2^{(i)}, \dots, \mathbf{r}_{m_i}^{(i)}$ are m_i L.I. eigenvectors corresponding to λ_i .

Case 2

By Theorem 2.6, there are m_i L.I. generalized eigenvectors for λ_i : $\xi_1^{(i)}, \xi_2^{(i)}, \dots, \xi_{m_i}^{(i)}$. We use each of them to construct a solution. For the j -th generalized eigenvector $\xi_j^{(i)}$, we first construct the following m_i vectors:

$$\begin{aligned} \eta_0^{(i,j)} &:= \xi_j^{(i)}, \\ \eta_1^{(i,j)} &:= (\mathbf{A} - \lambda_i \mathbf{I}) \xi_j^{(i)}, \\ \eta_2^{(i,j)} &:= (\mathbf{A} - \lambda_i \mathbf{I})^2 \xi_j^{(i)}, \\ &\vdots \\ \eta_{m_i-1}^{(i,j)} &:= (\mathbf{A} - \lambda_i \mathbf{I})^{m_i-1} \xi_j^{(i)}. \end{aligned}$$

Remark. Some of these vectors may be zero, because if $\xi_j^{(i)}$ is of order k ($k \leq m_i$) then $\eta_k^{(i,j)} = \eta_{k+1}^{(i,j)} = \dots = \mathbf{0}$, and $\eta_{k-1}^{(i,j)}$ is an ordinary eigenvector, while the rest $k - 1$ nonzero vectors are generalized eigenvectors.

Now we construct the solution as:

$$\mathbf{y}_j^{(i)}(t) = \left(\eta_0^{(i,j)} + \eta_1^{(i,j)} t + \eta_2^{(i,j)} \frac{t^2}{2!} + \dots + \eta_{m_i-1}^{(i,j)} \frac{t^{m_i-1}}{(m_i-1)!} \right) \cdot e^{\lambda_i t} \quad \text{for } j = 1, 2, \dots, m_i.$$

It can be verified that $\mathbf{y}_j^{(i)}$ is indeed a solution to the system by direct substitution: $\frac{d}{dt}(\mathbf{y}_j^{(i)}) = \mathbf{A} \cdot \mathbf{y}_j^{(i)}$.

The General Solution

Finally, we can express the general solution to the system as

$$\begin{aligned} \mathbf{y}(t) &= \sum_{i=1}^k \sum_{j=1}^{m_i} c_j^{(i)} \mathbf{y}_j^{(i)}(t) \\ &= \sum_{i=1}^k \sum_{j=1}^{m_i} c_j^{(i)} \left(\eta_0^{(i,j)} + \eta_1^{(i,j)} t + \eta_2^{(i,j)} \frac{t^2}{2!} + \dots + \eta_{m_i-1}^{(i,j)} \frac{t^{m_i-1}}{(m_i-1)!} \right) \cdot e^{\lambda_i t}. \end{aligned}$$

3 Non-Homogeneous Systems with Constant Coefficients

3.1 Method of Undetermined Coefficients

3.2 Variation of Parameters